



Characterizing the Histology Spatial Intersections Between Tumor-Infiltrating Lymphocytes and Tumors for Survival Prediction of Cancers Via Graph Contrastive Learning

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Abstract. Tumor-infiltrating lymphocytes (TILs) and their spatial interactions with tumors on whole-slide images (WSIs) of histopathology sections can provide valuable information about the immune response within the tumor micro-environment that is closely associated with the progression of human cancers. To effectively exploit the interactions between TILs and Tumors from WSIs, spatially informed analysis tools are required. Here, we present GCTIL, a simple but effective graph contrastive learning framework to learn meaningful representations for the TILs and tumor nodes extracted from the WSIs. Specifically, GCTIL considers the graph permutation of different strength to help learn robust node representations that can not only capture the structural characteristics of the graph but also preserve the correct distance orders among different permutations. Moreover, GCTIL also imposes distance constraints to distinguish the node embeddings of different types (i.e., TILs and Tumor). Then, based on the patch representation derived from GCTILs, we apply the graph attention networks (GATs) to describe the spatial interactions between TILs and tumor regions in WSIs for survival analysis of human cancers. We evaluate the performance of our method on the Breast Invasive Carcinoma (i.e., BRCA) cohort derived from The Cancer Genome Atlas (TCGA), and the experimental results indicate that our method is superior to the comparing methods.

Keywords: Graph Contrastive Learning · Survival Analysis · Tumor-Infiltrating Lymphocytes · Whole-Slide Histopathological Image

1 Introduction

The tumor micro-environment (TME) is a complex ecosystem composed of not only tumor cells but also non-cancerous parts, including immune cells, fibroblasts, endothelial cells, and extracellular matrix components [9]. Among various

non-cancerous components, the Tumor-infiltrating lymphocytes (TILs) plays an important role in host immune system, which is crucial in regulating tumor growth by continuous immunosurveillance and initiation of inflammatory reactions [2].

With the recent advent of cost-effective whole-slide digital scanners [18], it has become the research hot-spot to analyze the immune mechanism of TILs based on the digital whole-slide histopathological images (WSIs). For instance, Wang et al. [16] propose a computational pipeline that can spatially characterize the TILs in tumor specimens to predict response to immune checkpoint inhibitors in patients with gynecological cancer. Saltz et al. [9] present the spatial TILs maps for 13 cancer cohorts, and validate that the distribution of TILs is closely associated with the prognosis of human cancers. All these studies suggest that analyzing TILs based on the WSIs can provide more valuable insights into the spatial distribution of lymphocytes within the tissues than the approaches relying on the patients' genomic profile.

Although much progress has been achieved, most of the existing WSI-based studies only focus on analyzing the spatial distribution of TILs, which overlooks exploring the interactions between TILs and tumors that can provide valuable insights into the body's immune response to human cancer [10]. In order to characterize the spatial associations among different patches in WSIs, several graph convolutional network (GNN) based studies are proposed [3, 6]. However, these GNN based studies have the following two limitations. Firstly, these studies only consider the spatial relationship among different tumor patches, and thus neglect to take the Tumor-TIL interaction information affecting patients' survival into consideration. Secondly, to the best of our knowledge, all the existing WSI-based graph learning algorithms directly apply the pre-trained network *e.g.*, ResNet101 [6], VGG16 [3] on the selected patches to extract node features. However, such methods have difficulty in extracting meaningful node-level features representing the structure of the graph.

To address the above limitations, in this paper, we propose a novel graph contrastive learning framework *i.e.*, GCTIL to learn meaningful representations for the TILs and tumor nodes extracted from the WSIs. GCTIL considers the graph permutation of different strength to help learn robust node representations that can not only capture the structural characteristics of the graph but also preserve the correct distance orders among different permutations. Then, based on the learned features of different patches (nodes), the graph attention network interleaved with the graph pooling layer is applied to characterize the interaction strength between TILs and tumor patches for the prognosis prediction of human cancers. We evaluate the performance of our method on the Breast Invasive Carcinoma (*i.e.*, BRCA) cohort derived from The Cancer Genome Atlas (TCGA), the experimental results indicate that our method can achieve better prognosis performance than its competitors (Fig. 1).

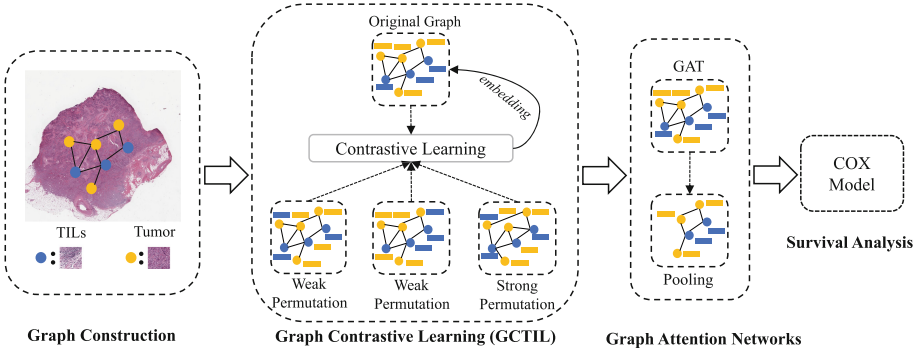


Fig. 1. The flowchart of the proposed method.

Table 1. Demographics information for the BRCA cohort in TCGA.

Characteristics	Summary
Patients:	
Censored	627
Non-censored	63
Age (Y):	54.6
Follow-up (M):	25.3

2 Materials and Methods

The flowchart in this study consists of three steps including graph construction from WSIs, developing GCTIL for node representation learning and applying graph attention networks (GATs) for survival analysis of human cancers.

Dataset: In this study, we test the performance of our method (*i.e.*, GCTIL) on the Breast Invasive Carcinoma (BRCA) cohort derived from the Cancer Genome Atlas (TCGA) [8]. We show the demographic information of BRCA cohort in Table 1, where the censored patients refer to the individuals who have not experienced the death events while the non-censored category correlate to the patients whose survival times are the exact time from initial diagnosis to death.

Graph Construction from WSIs: Since each of the collected Whole-slide pathological images (WSI) is extremely large in size (*e.g.*, 100,000-by-100,000 pixels), We first divide it into non-overlap patches with the size of 512×512 . Then, considering that the generated patches may lack enough tissues for further analysis, we select the patches whose image density is larger than 0.7 as candidate patches [20]. Here, the image density is calculated as the percentage of non-white (at least one of the red, green, and blue values is below 200 in the 24-bit RGB color space) pixels in that patch. Next, based on the annotated dataset

released by [1], we train a U-net++ network for the semantic segmentation of tumor, TILs regions in each patch. After getting the pixel-level segmentation results of each valid patch, we calculate its tumor, lymphocytes area ratios and select 300 patches with the largest ratios for each patch type. Based on the extracted tumor and TILs patches, we construct a graph for each WSI. Here, given patches as nodes, we adopt the K-Nearest Neighbors (KNN) algorithm to construct the graph based on the Euclidean distances between patch pairs, and the number of nearest neighbors for each node is set to 50.

Graph Contrastive Learning for the Representations of Tumor and TILs Patches: For the extracted patches derived from the WSIs, most of the existing studies usually apply the pre-trained network (*e.g.*, ResNet-101 and VGG-16) to acquire the node representations in the graph [22]. However, such representations cannot help learn robust representations characterizing the structure of the graph. Based on the above considerations, we have proposed a novel graph contrastive learning framework (*i.e.*, GCTIL) to learn node representations for different types of nodes (*i.e.*, tumor and TILs patches). Suppose that we are provided with a graph g_o that consists of a set of node features $g_o = \{x_o^1, x_o^2, \dots, x_o^N\}$, where N represents the number of nodes in the graph and $x_o^i \in R^F$ denotes the feature for node i . We are also provided with relational information between these nodes in the form of an adjacency matrix, $A \in R^{N \times N}$, and our goal is to learn an encoder $\varepsilon : R^{N \times F} \times R^{N \times N} \rightarrow R^{N \times Q}$, so that we can derive the high-level representation of the input node $H_o = \{h_o^1, h_o^2, \dots, h_o^N\}$, where $h_o^i \in R^Q$.

More specifically, the proposed GCTIL is composed of three parts. Firstly, GCTIL applies the noise-contrastive type objective (*i.e.*, Deep Graph Infomax [15]) to ensure that the learned node representations can capture the global structure information of the entire graph represented by the summary vector $u = \frac{1}{N} \sum_{i=1}^N h_o^i$. Based on g_o , we also define its three variations with different permutation strength. Specifically, the two weak permutations of the original graph *i.e.*, g_{w1} and g_{w2} can be obtained by shuffling the nodes feature in g_o . Based on the weak permutation graph, we also generate one graph with strong permutation *i.e.*, g_s by additionally shuffling the adjacent matrix in g_o . Then, the noise-contrastive learning based objective function can be denoted as:

$$\begin{aligned}
 Loss_{GI} = & -\frac{1}{N} \sum_{g_p \in \{g_{w1}, g_{w2}, g_s\}} \left(\sum_{i=1}^N E_{g_o} [\log D(h_{g_o}^i, u)] \right. \\
 & \left. + \sum_{i=1}^N E_{g_p} [\log(1 - D(h_{g_p}^i, u))] \right)
 \end{aligned} \tag{1}$$

where $h_{g_p}^i$ refers to the output of node i for permutation $g_p \in \{g_{w1}, g_{w2}, g_s\}$, and D is a discriminator function. By applying such binary cross-entropy (BCE)

loss between the original graph and its permutations, we are able to learn node representations that can effectively capture the global information of the graph.

In addition, in order to learn the robust node representations from the graph, we design a quadruplet loss which further constrains the distance among the node representations of the input graph and its variations with different permutation strength. Specifically, on one hand, by considering that the graph with strong permutation is influenced more than that with weak permutation, we add the following loss function to enforce their node representations:

$$L_{ow} = \frac{1}{N} \sum_{i=1}^N \max(0, d_{o,w_1}^i - d_{o,s}^i + \alpha) \quad (2)$$

where $d_{o,w_1}^i = \|h_o^i - h_{w_1}^i\|_2$, $d_{o,s}^i = \|h_o^i - h_s^i\|_2$ and α is the margin variable. Intuitively, Eq. (2) enforces that the node distance between the original graph and its weak permutation should be smaller than that between the input graph and its strong permutation. On the other hand, we also notice that the variations among the nodes representations of different weak permutations should be smaller than that between the original graph and its strong permutation. Hence, we additionally design the following loss function:

$$L_{ww} = \frac{1}{N} \sum_{i=1}^N \max(0, d_{w_1,w_2}^i - d_{o,s}^i + \beta) \quad (3)$$

where $d_{w_1,w_2}^i = \|h_{w_1}^i - h_{w_2}^i\|_2$ and β is the margin variable. By the combination of Eq. (2) and Eq. (3), we can derive the second part of the objective function of GCTIL as:

$$Loss_P = L_{ow} + L_{ww} \quad (4)$$

Finally, since the patches with different types (*i.e.*, tumor and lymphocytes) used in this study usually show diverse appearance and arrangement, we also impose distance constraints to distinguish the node embeddings of different types. Here, suppose the sets of features for tumor and lymphocytes nodes in g_o are denoted as $(H_o)^T = \{(h_o^1)^T, (h_o^2)^T, \dots, (h_o^{N_T})^T\}$ and $(H_o)^L = \{(h_o^1)^L, (h_o^2)^L, \dots, (h_o^{N_L})^L\}$, respectively, where N_T and N_L represent the number of tumor and lymphocytes nodes. Then, we define a Maximum Mean Discrepancy (MMD) based loss function to enlarge the distance among them:

$$Loss_C = - \left\| \frac{1}{N_T} \sum_{i=1}^{N_T} (h_o^i)^T - \frac{1}{N_L} \sum_{j=1}^{N_L} (h_o^j)^L \right\|_H \quad (5)$$

where H is the Hilbert space. By the combination of Eq. (1), Eq. (4) and Eq. (5), the main objective function of the proposed GCTIL is formulated as:

$$Loss_G = \lambda \times Loss_{GI} + \mu \times Loss_P + Loss_C \quad (6)$$

where λ and μ are regularization parameters that need to be tuned during the cross-validation procedure.

Calculating TILs-Tumor Interaction via Graph Attention Network:

After deriving the representations of different types of nodes, we employ the graph attention networks (GATs) to characterize the interaction between different TILs and tumor patches. The input of the GAT layer is $H_o = \{h_o^1, h_o^2, \dots, h_o^N\}$ which represents the output features of the proposed GCTIL method, where $h_o^i \in R^Q$. The GAT layer generates a new group of node features $H'_o = \{\bar{h}_o^1, \bar{h}_o^2, \dots, \bar{h}_o^N\}$ via a weight matrix $W \in R^{Q \times Q'}$, where $\bar{h}_o^i \in R^{Q'}$. Next, with a shared attentional mechanism $R^{Q'} \times R^{Q'} \rightarrow R$, we calculate the attention coefficients among different nodes, which can be formulated as:

$$e_{ij} = a \left([W\bar{h}_o^i \| W\bar{h}_o^j] \right), j \in \mathcal{N}_i. \quad (7)$$

Furthermore, a softmax function is then adopted to normalize the attention coefficients denoted by α_{ij} . After that, the new feature vector $\bar{h}_o^i$ for node i is calculated via a weighted sum:

$$\bar{h}_o^i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} W h_o^j \right). \quad (8)$$

where $\mathcal{N}_i$ represents all neighbors of node i . In addition, in order to avoid over-fitting and identify survival-associated interactions among different types of TME components, we adopt SAGPool [5] to select top s ($s \in [0, 100]$) percentage of nodes with high attention scores. Finally, at the last layer of GAT, a global attention pooling strategy introduced in [7] is applied to aggregate the node features into WSI-level feature for the following survival analysis task.

Survival Analysis: After deriving the WSI-level representation for M patients $[f(W_1), f(W_2), \dots, f(W_M)]$, we apply the widely used Cox proportional hazard model [14] to conduct the survival analysis of human cancers. In order to evaluate the prognosis performance of different methods, the Concordance Index (CI) is applied [11]. The CI values range from 0 to 1, and the larger values indicate better prognosis performance and vice versa.

3 Experiment

Experimental Settings: We apply the 5-fold cross validation strategy to evaluate the performance of different methods. In the training dataset, 25% of the samples are randomly chosen to form the validation set for parameters tuning. For the proposed framework of GCTIL, we firstly utilize the pre-trained ResNet-101 to extract features from the derived patches as the input features. The embedding network of GCTIL consists of three graph convolution layers with the dimensions of 128. The parameters λ and μ in Eq. (6) are tuned from 0.8 to 1.2 with the interval 0.1, while the parameters α and β in Eq. (2) and Eq. (3) are both tuned from 0.1 to 0.3 with the interval 0.1. The Graph Attention Networks (GATs) introduced in this study is consisted of three GAT layers interleaved with 2 graph pooling layers, the percentage of the preserved nodes in each graph pooling layer is tuned from 0.3 to 1 with the interval 0.1.

Table 2. Comparisons of Our Method with Other Models by the Measurements of CI

Method	CI	Method	CI
DeepGraphSurv [6]	0.614 ± 0.05	DeepAttnMISL [19]	0.598 ± 0.04
GRACE [21]	0.621 ± 0.05	MVGRL [4]	0.624 ± 0.04
BGRL [13]	0.652 ± 0.02	CluSiam [17]	0.651 ± 0.07
GCTIL-N1	0.654 ± 0.02	GCTIL-N2	0.663 ± 0.03
GCTIL-N3	0.656 ± 0.03	GCTIL(Our Method)	0.682 ± 0.02

Comparison of GCTIL with other Survival Analysis Models: In this section, we compare our GCTIL with the following nine methods: (1) DeepGraphSurv [6]: A graph convolutional neural network based survival prediction model. (2) DeepAttnMISL [19]: A deep multiple instance learning based survival prediction model. (3) GRACE [21]: A graph contrastive learning framework for acquiring node representation. (4) MVGRL [4]: A self-supervised learning approach for node representation learning. (5) BGRL [13]: A graph representation learning method based on predicting the alternative augmentations of the input. (6) CluSiam [17]: A self-supervised learning (SSL) based representation learning method for survival prediction from WSIs. 7) GCTIL-N1: A variant of GCTIL without the noise-contrastive loss ($Loss_{GI}$ in Eq. (6)). 8) GCTIL-N2: A variant of GCTIL without the quadruplet loss ($Loss_P$ in Eq. (6)). 9) GCTIL-N3: A variant of GCTIL without distinguishing different types of nodes ($Loss_C$ in Eq. (6)). As indicated in Table 2, firstly, we observe that applying the contrastive learning (*i.e.*, GRACE, GCTIL) or the self-supervised learning algorithms (*i.e.*, MVGRLH, BGRL, CluSiam) to learn the encodings of the graph nodes can more improve the prognosis results than the methods simply applying the pre-trained network (*i.e.*, DeepGraphSurv, DeepAttnMISL) to acquire the node representations. This is because most of these node representation algorithms considered the topological organization of the patches derived from the whole-slide pathological images that are closely associated with cancer progression [12]. Secondly, our proposed GCTIL is significantly better than the competing methods. The reason lies in the fact that besides capturing the structural information of the graph like the existing studies, we also consider the distance ordering information among the permutation of different strength and the node type difference. Finally, we observe that the proposed GCTIL can achieve higher CI values than its variants (*i.e.*, GCTIL-N1, GCTIL-N2, GCTIL-N3), indicating the importance of each regularization term in the objective function of our GCTIL.

Comparison of GCTIL with Other Methods for Patient Stratification:

For the prognosis prediction of cancer patients, another important task is to stratify cancer patients into different survival subgroups, by which we could more precisely implement personalized treatment for them. In this study, we follow the method introduced in [11] applying the K-means clustering algorithm to separate the WSI-level representations of different patients in the testing set into

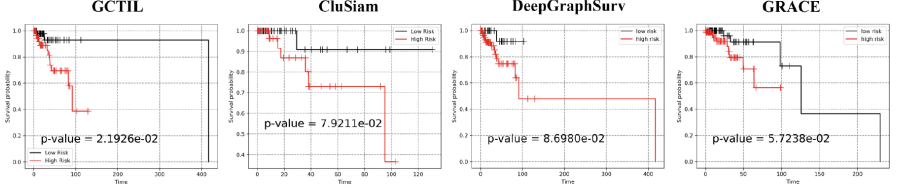


Fig. 2. Comparison of the Stratification Performance Among Different Methods.

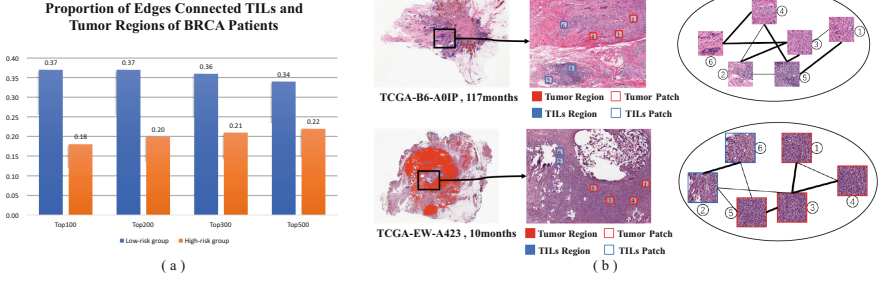


Fig. 3. (a) Proportion of Edges Connected TILs and Tumor patches among high and low survival-risk patients. (b) Visualization of the connections between TILs and tumor patches.

2 groups. Finally, we test if these two groups have distinct survival times using the Kaplan-Meier (KM) estimator and log-rank test [11]. In the log-rank test, the better patient stratification performance usually comes with smaller p-values and vice versa. As can be seen from Fig. 2, the proposed GCTIL can achieve the p-values of 2.19×10^{-2} that is smaller than the comparing methods. These results again validate the advantages of our GCTIL for the survival prediction of cancers.

Analyzing the Interactions Between Tumor and Lymphocytes Patches:

since the proposed graph attention networks are interleaved with the graph pooling layers that can select important nodes and edges that are highly correlated with patients' survival. Hence, for each patient, we select the top 100, 200, 300, 500 edges with the largest weights, and calculate the proportion of edges that connect TILs and tumor patches for the high and low survival-risk patients, respectively. As can be seen in Fig. 3(a), it is obvious that the proportions of edges connecting Tumor and TILs patches for low survival-risk patients are higher than that in high survival-risk patients. Such observations is consistent with the existing knowledge that the rich interactions between tumor and lymphocytes will stimulate the host immune response that could control the tumor growth and reduce survival risks [9]. Moreover, we also visualize the sampled tumor and lymphocytes patches and their connected edges in Fig. 3(b). Here, the thick black lines refer to the edges with higher weights calculated via GATs. Obviously, the weights and the number of the edges connecting tumor and TIL regions are higher for patients in low-survival risk group. These observations further highlights the advantage of

our GCTIL in providing in-depth explanations for the mechanism of the cancer development.

4 Conclusion

In this paper, we propose a graph contrastive learning framework *i.e.*, GCTIL to learn meaningful representation of Tumor and TILs patches in the whole-slide histopathological images (WSIs) for the survival analysis of human cancers. GCTIL considers the permutations of different strength to learn robust embeddings of different node types which can capture the global structure of the graph. Extensive experimental results on Breast Invasive Carcinoma cohort derived from the TCGA dataset demonstrate that GCTIL can not only achieves higher prognosis results but also has good explainable ability.

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